Communication

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Forensic Analysis Platform
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	**Team Standup**	Daily	Slack / Google Meet	Core Development Team (Rahul Rajak & Team)	Review daily sprint goals, model pipeline progress (RT-DETR, Whisper, OpenCV), and active task blockers.
2	**Progress Update**	Weekly	GitHub Projects & Email	Project Mentors & Evaluators	Report milestone completion, performance metrics, and deliverable review status.
3	**Issue / Bug Discussion**	As Needed	GitHub Issues & Discord	Developers & AI Engineers	Troubleshoot cloud deployment memory limits, CORS policies, WebGL 3D sync, and audio conversion bugs.
4	**Stakeholder Review**	Bi-Weekly	Google Meet / Live Demo	Academic Mentors & Industry Evaluators	Present forensic decision-support features, court-admissible PDF reports, and UI usability.
5	**Final Demo Rehearsal**	Once	Screen Share Session	Entire Team & Presenter	End-to-end dry run of the 10-minute multimodal demonstration covering live video, audio, and image workflows.

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Render cloud hosting free-tier 512MB RAM limitation caused out-of-memory container terminations when importing large neural network models.	Switched to CPU-only PyTorch build, lightweight OpenCV contour tracking, Whisper Base model, and implemented defensive fallback data responses.
2	Cross-Origin Resource Sharing (CORS) and path mismatch between Vercel frontend deployment and Render backend gateway.	Configured Flask-CORS (origins='*'), standardized unified /api/ endpoints, and synchronized static media assets directly to Vercel.
3	Coordinating synchronous temporal playback across 2D CCTV video player and 3D spatial coordinate projection in Three.js.	Implemented unified HTML5 player time-update listeners driving 3D suspect trajectory coordinates at 1Hz synchronized intervals.